Fabricating Amorphous Single-Phase Complex Concentrated Oxides for CO_2 Reduction

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Introduction

In 2024, **carbon dioxide** (CO_2) emissions reached a record high of 37.8 Gt CO_2, with an increase primarily caused by consumption of natural gas and coal.[1] Converting CO_2 into value-added, **single-carbon-containing products** presents an economically advantageous opportunity to achieve a sustainable carbon cycle and combat climate change.

Background

- We can use **amorphous** complex concentrated oxide (CCO) electrocatalysts with good **activity** and **selectivity** to effectively convert CO_2.
- 3 Component (Comp.) InZnSn-based oxides are shown to have good selectivity toward **formic acid**.[2]
- However, they are relatively **expensive** due to having a large indium composition.[2]

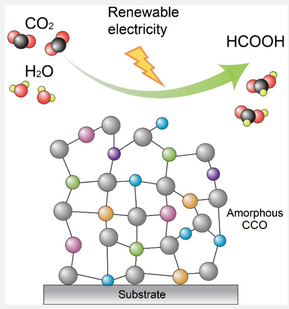


Fig 1. CO_2RR occurring on the surface of a CCO, reducing CO_2 to formic acid.[4]

Procedure

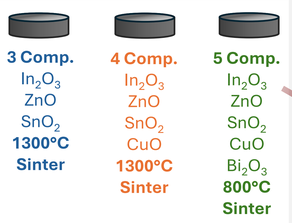


Fig 2.1. Bulk 3, 4, and 5 Comp. ceramics used for **PLD**.[3]

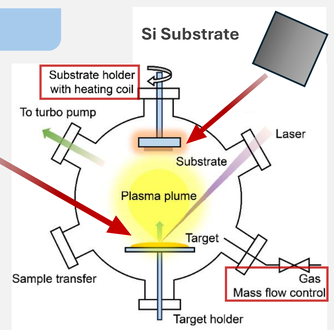


Fig 2.2. Adjust **temperature** and **oxygen partial pressure** to make uniform, amorphous, single-phase thin films using PLD.[4]

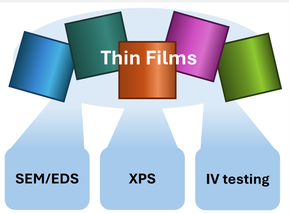


Fig 2.3. **Thin films** are characterized using:

SEM/EDS: Surface morphology, Elemental composition & distribution.

XPS: Vacancy concentrations.

IV testing: Finding conductivity (σ).

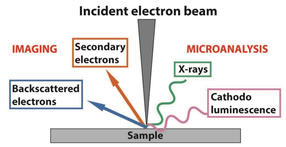
Samples Used

CCO Type	Deposition Temperatures (°C)	Pressure (mTorr)
3 Comp.	25, 100, 200, 300, 400, 500, 600, 700	0.607
4 Comp.	100, 200, 300, 400, 500, 600	0.607
5 Comp.	25, 100, 200, 300, 400, 500, 600, 700	0.607
4 Comp.	400	0.015, 0.193, 0.311
5 Comp.	300, 400	0.453

Results

Scanning Electron Microscopy (SEM) and Energy Dispersive X-ray Spectroscopy (EDS) confirm Pulsed Laser Deposition (PLD) is a successful technique for fabricating uniform, single-phase CCOs.

Fig 3.1. Types of signal from scattering in SEM/EDS.[5]



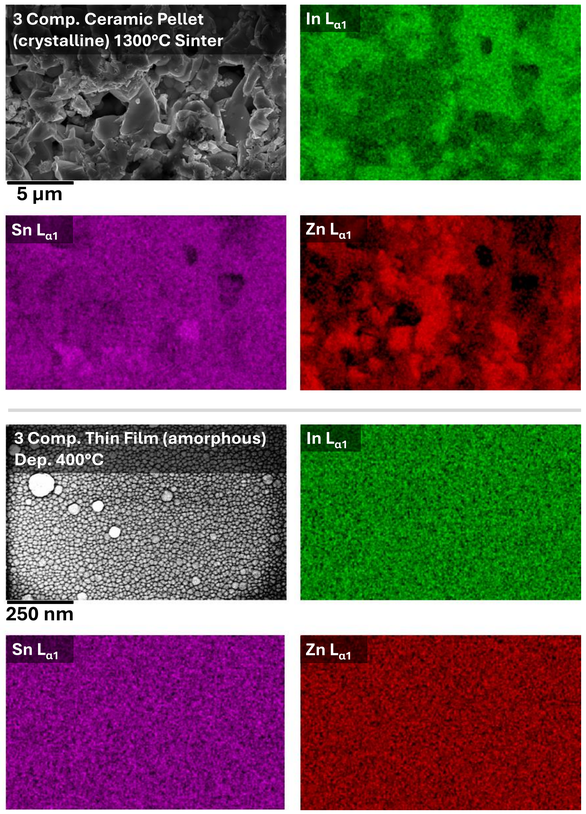
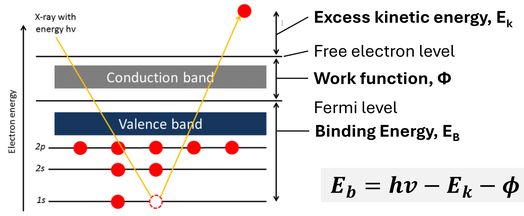


Fig 3.2. (top) 3 Comp. ceramic pellet SEM/EDS mapping shows non-uniform elemental distribution and **(bottom)** thin film is uniform. **XRD results support that thin films are amorphous and single-phase.**[4]

X-ray Photoelectron Spectroscopy (XPS) shows decreasing vacancy concentration with thin film crystallization.



$$E_b = h\nu - E_k - \phi$$

Fig 4.1. How binding energy is determined in XPS measurements.[6]

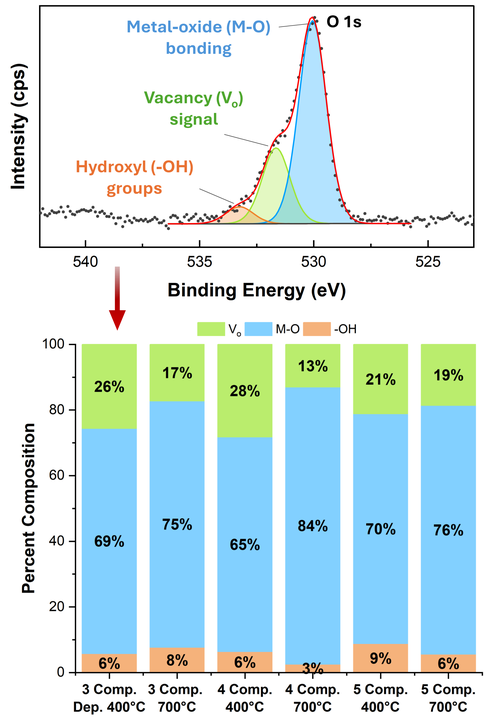
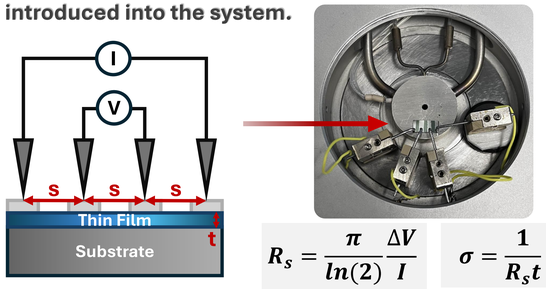


Fig 4.2. (top) XPS plot of the oxygen 1s orbital. **(bottom)** Percentage of vacancies in the thin film is related to surface chemical environment.

4-probe Current-Voltage (IV) testing results in decreasing conductivity as additional cations are introduced into the system.



$$R_s = \frac{\pi}{ln(2)}\frac{\Delta V}{I} \qquad \sigma = \frac{1}{R_s t}$$

Fig 5.1. (left) Depiction of the 4-probe IV test **(right)** Photograph of the experimental setup.

$$O_o^{\times} \rightarrow V_o^{\cdot\cdot} + 2e' + \frac{1}{2}O_2$$

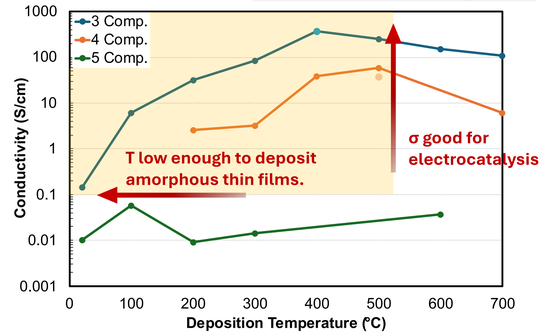


Fig 5.2. Conductivity of 3, 4, & 5 Comp. samples prior to the program.[4] **Corroborated data from the updated IV testing setup reinforces the result.**

Conclusion

- **Single-phase, uniform, amorphous** thin film CCOs can be fabricated using PLD.
- **Vacancy concentrations decrease as thin films crystallize,** lowering conductivity.
- 3 & 4-Comp. thin films have **highest conductivity** near 400-500°C **before crystallization**. 4-Comp. maintains a **high conductivity without needing as much indium**.
- 3 & 4-Comp thin films are good candidates for electrocatalysis, **5-Comp. films are not**.

Future Directions

- **Reduced oxygen partial pressure thin films have been prepared**; investigate effects of vacancy concentrations, independent of changes in deposition temperature.
- Obtain a value with error for the resistivity of 5-Comp. samples using the cross-plane IV testing.
- For samples that are good candidates for electrocatalysis, selectivity will be measured by collaborators.

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Citations

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[6] A. Vandenbroucke. (2015) Thesis.

Supplemental Information Available: